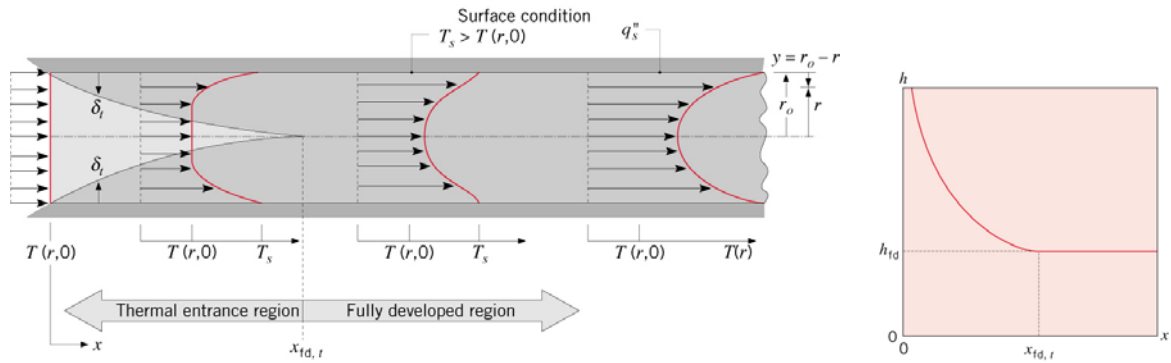


16. Thermally fully developed (FD) flow



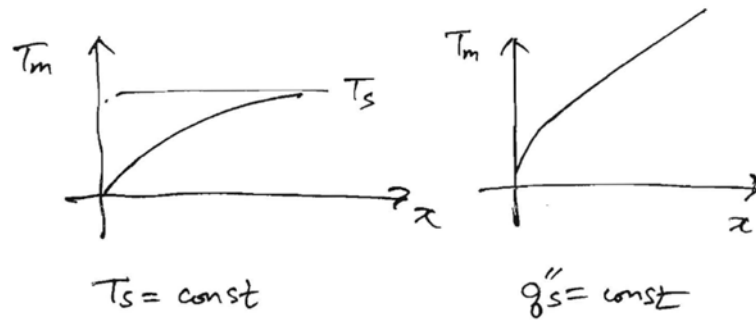
In thermally FD region, T varies with x but

- $h = \text{const} \leftarrow$ Definition of thermally fully developed flow
- Equivalently, dimensionless temperature $\frac{T - T_s}{T_m - T_s} = \text{const} \leftarrow$ Def. of thermally FDF

flow

$$\rightarrow \frac{\partial}{\partial x} \left[\frac{T_s(x) - T(r, x)}{T_s(x) - T_m(x)} \right]_{fd,t} = 0 \quad \text{or} \quad \frac{T_s(x) - T(r, x)}{T_s(x) - T_m(x)} \neq f(x)$$

Note $\frac{\partial T(r, x)}{\partial x} \neq 0$ and $\frac{dT_m(x)}{dx} \neq 0$ as shown below



$$\begin{aligned} \because \frac{T_s - T}{T_s - T_m} \neq f(x) &\rightarrow \frac{\partial}{\partial r} \left(\frac{T_s - T}{T_s - T_m} \right) \neq f(x) \rightarrow \frac{-\frac{\partial T}{\partial r}}{T_s - T_m} \neq f(x) \rightarrow \frac{-\frac{\partial T}{\partial r} \Big|_{r=r_0}}{T_s - T_m} \neq f(x) \\ &\rightarrow \frac{k \frac{\partial T}{\partial r} \Big|_{r=r_0}}{T_s - T_m} \neq f(x) \rightarrow \frac{q_s''}{T_s - T_m} \neq f(x) \rightarrow h \neq f(x) \end{aligned}$$

- Thermal entry length: $(x_{fd,t} / D)_{lam} \approx 0.05 \text{Re}_D \text{Pr}$
 $(x_{fd,t} / D)_{turb} = 10$

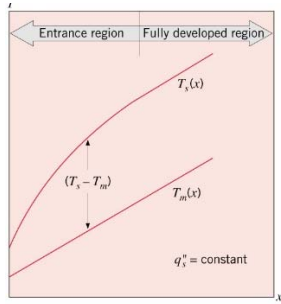
← These correlations are from numerical analysis / experiments

[Special cases]

Consider two typical boundary conditions

(I) $q_s'' = \text{const}$

$$q_s'' = h(T_s - T_m) = \text{const} \rightarrow \left. \frac{dT_s}{dx} \right|_{fd,t} = \left. \frac{dT_m}{dx} \right|_{fd,t}$$



(II) $T_s = \text{const}$

$$\frac{\partial}{\partial x} \left(\frac{T_s - T}{T_s - T_m} \right) = 0 \rightarrow \frac{\partial T}{\partial x} = \frac{dT_s}{dx} - \frac{T_s - T}{T_s - T_m} \frac{dT_s}{dx} + \frac{T_s - T}{T_s - T_m} \frac{dT_m}{dx}$$

$$\rightarrow \left. \frac{\partial T}{\partial x} \right|_{fd,t} = \left. \frac{T_s - T}{T_s - T_m} \frac{dT_m}{dx} \right|_{fd,t}$$

